SpheroidMetrics

Automated Segmentation and Morphometric
Analysis of Bright-Field Images Cancer Spheroi

Key stakeholders

Client

Dr. Dalia Mahdy

Project supervisor

Prof. Dr. Magda Gregorová

Team

Tulasi Rajgopal

Sara Prasla

Richard

Background

Three-dimensional (3D) cell cultures such as spheroids are widely used in-vitro models to study tumor growth, drug response, and cell behavior as they closely mimic the structure and microenvironment of real tissues. Monitoring changes in spheroid morphology such as size, shape, and brightness over time provides valuable insights into growth dynamics, cell proliferation, viability, and treatment effects. Accurate and efficient analysis of these features is essential for meaningful evaluation of spheroid behavior.

Problem statement

Manual measurement of spheroid features from microscopy images is slow and subjective, making it difficult to efficiently analyze large datasets or track changes over time. Therefore, there is a need for automated image segmentation and quantitative analysis that is both faster and more reliable, enabling accurate measurement of morphological parameters such as area, perimeter, roundness, and brightness across different time points.

Objective

This project aims to develop an automated image-analysis pipeline to segment spheroids in bright-field images, extract morphological and intensity metrics, and compare them to assess growth and drug-induced changes.

Objective

1 Image Segmentation

Implement and optimize a segmentation method to accurately detect spheroid boundaries from bright-field microscope images (d7 and d10).

2 Feature Extraction

Calculate key morphological metrics for each spheroid:

- Area
- Perimeter
- Roundness / Circularity

Extract raw pixel intensity values (brightness) within the spheroid region.

3 Feature Comparison

Perform comparative analysis of extracted metrics between images taken before and after drug treatment.

Project roadmap (High level plan)

thws

6th oct | 11th Nov

21st Oct| 17th Nov

14th Nov | 26st Nov

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Research and Planning

1. Find the related papers for different aspects of the project. -metrics, models,data and preprocessing.
2. Other resource gathering- finding tools and learning to use tools
3. Overall detailed project planning

Data collection and Annotation

1. Data collection
2. Manual annotation
3. Validation (by biologist)
4. Data organisation in different categories- same spheroid images to same class.
5. Bibin’s data(Back up/cross validation)???

Implementation & Training

1. Preprocessing of Data
2. Explore 2 models(yolo, U-net)
3. Sources/tools
4. Code Implementation
5. Models training & Validation
6. Compare & Validate with roboflow annotation- IoU,Dice

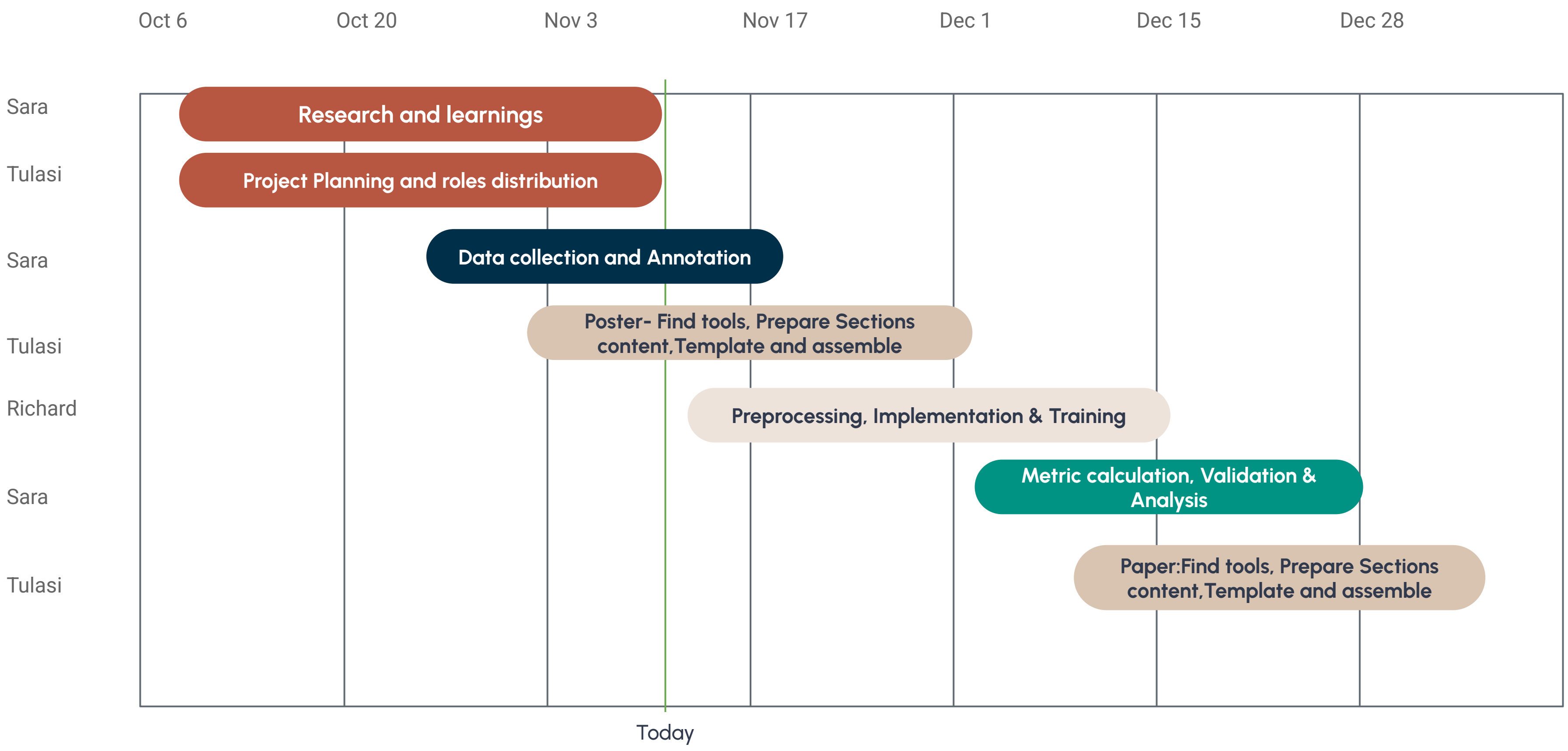
Metric calculation, Validation & Analysis

1. List of metrics
2. Software/scripts
3. Output extraction

Documentation(Poster+ Final Paper)

1. Find the suitable tools
2. Create methodology flowchart
3. Draft the design and template
4. Final Alignment and finishing

Project timeline (High level plan)



Deliverables

Deliverable	Due date	Accountable
Project plan	9/11	Tulasi
Annotated and Organised data	14/11	Sara
Poster	23/11	Tulasi
Segmentation Pipeline script with trained model (yolo)(U-net)	15/12	Richard
Metrics calculation	–	Sara
Paper	–	Tulasi

Roles and Responsibilities

- **Project Lead / Coordinator** → communication, timeline, reports - Tulasi
- **Technical Lead** → architecture, implementation oversight - Richard
- **Research Lead** → literature review, Data annotation, evaluation, experimentation - Sara

	Sara Prasla	Richard	Tulasi
Literature Review	R A	R	R
Planning and Task Creation	R	C	A R
Data collection and organising	R A	I	I
Implementation & Training	C	A R	R
Validation & Analysis	A R	R	R
Documentation(Poster + Final Paper)	C R	C R	R A

R	Responsible
A	Accountable
C	Consulted
I	Informed

Sources / Tools / Technology

1. Geometrical Metrics - Area, Circularity, Perimeter and Brightness
2. Models and it’s validation - YOLO, U-net, TransNext, Precision, Recall
3. Segmentation Validation - IoU/JCD, Dice
4. Tools/software - Git, Google drive, Roboflow, VS code, Key notes, Lucidchart

Meetings: Progress tracking

1. Internal meetings: weekly team stand-up (short progress check-ins)

    - **Every Tuesday at 10am**

2. Client meetings: weekly updates or milestone reviews

    - **Every Friday at 11am**

Risk Analysis

- **Time constraints** → Request Dalia to provide fully annotated data, and deliver incremental results while working diligently to meet all deadlines.
- **Lack of data** → Need to perform augmentation to increase the data.
- **Model Implementation** →Simultaneously test multiple models and hyperparameters on a small dataset first.
- **Validation and Metric Calculation** →Use multiple metrics and Visual inspection of segmented images for quality assurance.

Deliverables and Success Metrics

- **Tangible outputs:** software prototype(Pipeline), report, trained model, etc.
- **Performance metrics:** accuracy, efficiency, usability, etc.
- **Acceptance criteria:** achieving accuracy over 90% for the segmentation and successful metrics analysis.

Thank you